Tribhuvan University
Institute of Engineering
Purwanchal Campus
पूर्वाञ्चल क्याम्पस

Data Flow Diagram

Project: Decentralized Smart Power Grid

Department of Computer & Electronics Engineering

Course	Software Engineering (Practical)
Course code	ENCT 352
Team	440 V
Submitted on	July 13, 2026

Team Members

S.N.	Name	Roll / ID
1	PRATIMA CHAUDHARY	PUR080BCT064
2	RABIN KATTEL	PUR080BCT067
3	SABIN BASNET	PUR080BCT077
4	SARBESH TIMSINA	PUR080BCT082

Submitted to: Asst. Prof. Binay Lal Shrestha

Contents

1 Introduction

1.1 Purpose

This report presents the Data Flow Diagram (DFD) for the proposed **Decentralized Smart Power Grid** system. The DFD illustrates how energy consumption data moves through the system, how prepaid billing is processed using blockchain smart contracts, and how administrators and consumers interact with the system for monitoring and management.

1.2 Scope

The DFD covers the operational flow of the live system. It includes smart meter data collection, telemetry processing, blockchain-based billing, anomaly detection, dashboard updates, and user interactions. The development and training of machine learning models, smart contract development, and hardware prototyping are outside the scope of this DFD, as they are part of the development process rather than the runtime operation.

1.3 Definitions and Abbreviations

Term	**Meaning**
DFD	Data Flow Diagram
IoT	Internet of Things
ESP32	Microcontroller used as the smart meter node
API	Application Programming Interface
AMI	Advanced Metering Infrastructure
ML	Machine Learning
Smart Contract	Blockchain program that automates prepaid billing
Telemetry	Real-time energy usage data transmitted by the smart meter
Consumer	Registered electricity user of the system
Administrator	Utility operator responsible for monitoring the system

2 Project Domain Analysis

2.1 Domain Description

Electrical power distribution systems are responsible for delivering electricity from utility providers to consumers while monitoring energy consumption for billing purposes. Traditional systems rely on centralized metering infrastructure and manual verification of billing records. The proposed project introduces a decentralized approach that combines IoT smart meters, blockchain technology, and machine learning to provide transparent prepaid energy management, secure transaction records, and real-time monitoring.

2.2 Existing System

In conventional power distribution systems, smart meters or electricity meters send consumption data to a centralized utility database. Billing is generally performed using postpaid models, where consumers receive their bills after energy has been consumed. Consumers have limited visibility into how charges are calculated, and the centralized architecture creates a single point of failure for data storage and billing operations.

2.3 Proposed System

The proposed Decentralized Smart Power Grid continuously receives electricity consumption data from ESP32-based smart meter nodes. The backend validates the incoming telemetry and forwards billing requests to blockchain smart contracts that automatically deduct prepaid credit. Machine learning models analyze consumption patterns to estimate remaining credit duration and detect possible power theft or abnormal usage. The processed information is stored securely and displayed on a web dashboard, allowing both consumers and administrators to monitor energy usage, account balance, billing history, and system alerts in real time.

3 DFD Design

3.1 External Entities, Processes, and Data Stores

External Entities

- Consumer
- Smart Meter (ESP32)
- Grid Administrator

Main Processes

- Receive Meter Data
- Process Energy Consumption
- Execute Smart Contract Billing
- Detect Anomalies and Predict Usage
- Update Dashboard and Notifications

Data Stores

- D1 Consumer Database
- D2 Energy Usage Database
- D3 Blockchain Ledger
- D4 Machine Learning Models

3.2 Level 0 DFD

The Level 0 DFD presents the overall context of the Decentralized Smart Power Grid System. Smart meters provide energy consumption data to the system. Consumers recharge prepaid balances and view usage information, while administrators monitor the system and receive alerts.

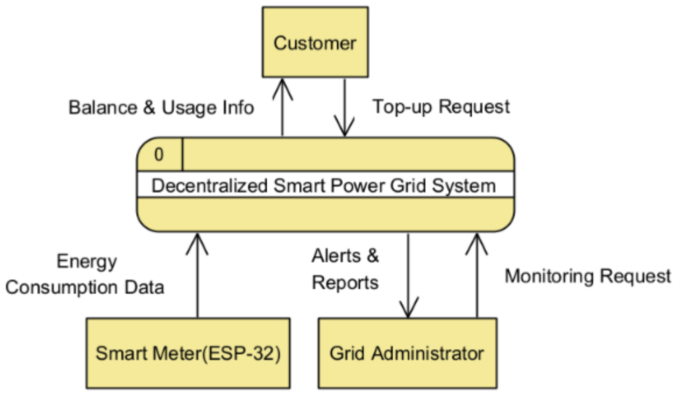

Figure 1: Level 0 DFD of the Decentralized Smart Power Grid System

3.3 Level 1 DFD

The Level 1 DFD expands the system into its five major processes: receiving telemetry data, processing energy consumption, executing blockchain-based billing, detecting anomalies and updating the dashboard.

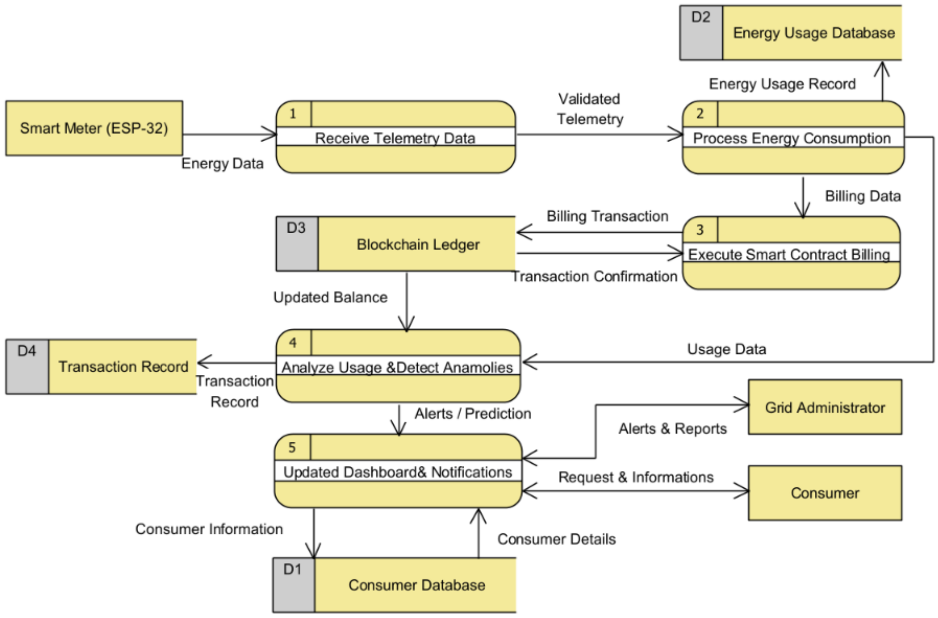

Figure 2: Level 1 DFD of the Decentralized Smart Power Grid System

3.4 Level 2 DFD for Execute Smart Contract Billing

The Level 2 DFD expands Process 3, Execute Smart Contract Billing. It verifies the consumer wallet, calculates energy charges, executes the blockchain smart contract, updates the remaining balance, and generates billing status.

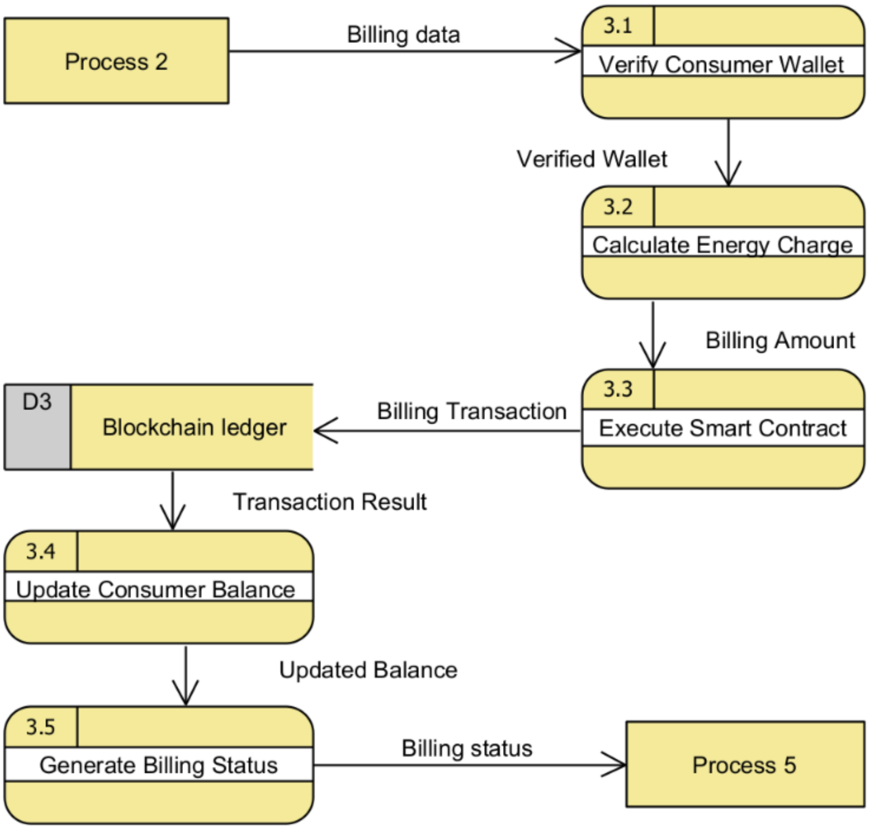

Figure 3: Level 2 DFD for Process 3: Execute Smart Contract Billing

3.5 Balancing

The Level 2 DFD maintains the same input and output as Process 3 in the Level 1 DFD. It receives billing data from Process 2 and returns updated balance and billing status to the following processes, ensuring proper balancing between the parent and child DFDs.

4 Requirements Derived from the DFD

4.1 Functional Requirements

ID	Functional Requirement	Priority
FR-1	The system shall receive real-time energy consumption data from ESP32-based smart meters.	High
FR-2	The system shall validate and process incoming telemetry data before billing.	High
FR-3	The system shall execute prepaid billing using blockchain smart contracts.	High
FR-4	The system shall detect abnormal energy consumption and possible power theft using machine learning models.	High
FR-5	The system shall update consumer balances, billing records, and transaction history after successful processing.	Medium
FR-6	The system shall provide dashboards for consumers and administrators to monitor energy usage, account status, and system alerts.	Medium

4.2 Non-functional Requirements

ID	Non-functional Requirement	Category
NFR-1	The system should process incoming telemetry with minimal delay to support near real-time monitoring.	Performance
NFR-2	The dashboard should provide an intuitive interface for viewing energy usage, prepaid balance, and billing history.	Usability
NFR-3	The system should securely store billing records and blockchain transaction references.	Security
NFR-4	The system should maintain reliable communication between smart meters, backend services, and blockchain nodes.	Reliability

5 Conclusion

This report presented the Data Flow Diagram (DFD) for the proposed Decentralized Smart Power Grid system. The diagrams describe how energy consumption data is collected from smart meters, processed by the backend, recorded through blockchain smart contracts, analyzed for anomalies, and presented to consumers and administrators.

The DFD provides a clear representation of the movement of data throughout the system and identifies the major processes, external entities, and data stores involved. It also serves as a foundation for the subsequent stages of software design, implementation, and testing by ensuring that the system's functional requirements are well understood.

6 References

1. IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications.
2. Smith, J., & Kumar, A. (2022). IoT-Based Smart Energy Meter for Real-Time Monitoring and Automated Billing. International Journal of Smart Grid Technologies, 10(3), 150-158.
3. J. Jithish, B. Alangot, N. Mahalingam, and K. S. Yeo, "Distributed Anomaly Detection in Smart Grids: A Federated Learning-Based Approach," IEEE Access, vol. 11, pp. 9824-9838, 2023.